Comparative Structural Analysis of the A-1 Chlorophyll Binding Pockets in Far-Red Photosystem I

Quantitative Microenvironment Characterization of Chl *a* (A-1A) vs. Chl *f* (A-1B) in *C. thermalis* Cryo-EM Structures (PDB: 9I9L & 9EYS)

INVESTIGATOR	TARGET COMPLEXES	ANALYSIS FOCUS	STATUS
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1. Executive Summary & Research Question

Oxygenic photosynthetic organisms adapted to far-red light (FaRLiP, Far-Red Light Photoacclimation) incorporate chlorophyll *f* (Chl *f*) to absorb photons beyond 700 nm. Within the Photosystem I (PSI) core reaction center, electron transfer proceeds along two pseudo-symmetric branches (A-branch and B-branch). Cryo-EM models reveal a striking site-specific asymmetry: Chl *f* is selectively incorporated at the primary electron acceptor **A-1B** site, while the pseudo-symmetric **A-1A** site retains standard chlorophyll *a* (Chl *a*).

The chemical distinction between Chl *a* and Chl *f* resides exclusively at the **C2 carbon position**: Chl *a* carries a methyl substituent (`-CH3` , CMB), whereas Chl *f* carries an electron-withdrawing formyl substituent (`-CHO` , CMB=OMB). This structural investigation asks: *Does the local protein microenvironment surrounding the C2 position provide distinct steric, hydrogen-bonding, or aromatic interactions that rationalize site-specific binding of Chl f at A-1B versus Chl a at A-1A?*

> **Key Finding Summary:** Across all six biological replicates (3 replicates in 9I9L and 3 replicates in 9EYS), the real formyl oxygen (OMB) at A-1B engages in **zero hydrogen bonds** (no polar donor/acceptor ≤ 3.3 Å). Instead, site selectivity is strongly correlated with **local aromatic stacking** (Phe459 at ~3.55 Å in A-1B vs Phe708 at ~5.64 Å in A-1A) and **differential packing density** (17–18 heavy atoms around C2 in A-1B vs 13–15 in A-1A).

2. Methodological & Computational Framework

All measurements were executed directly on atomic coordinates extracted from RCSB macromolecular CIF structures (`9i9l.cif` and `9eys.cif`) using BioPython's `MMCIFParser` and spatial indexing with KD-trees (`NeighborSearch`):

- **Spatial Site Identification:** Pigment clusters were identified using axial magnesium ligands. The Met-ligated chlorophylls (`A0` candidates) served as spatial anchors; non-Met chlorophylls with Mg–Mg distances ≤ 14.0 Å were cataloged as `A-1` candidates. Pairs were matched across branches: A-1B (formyl-bearing, F6C) and A-1A (non-formyl, CLA) separated by ~18.8 Å Mg–Mg distance.
- **Chemical Validation:** Unambiguous verification of the C2 substituent: Chl *a* contains CMB (methyl carbon, no oxygen bonded); Chl *f* contains CMB bonded to OMB (carbonyl oxygen, observed bond distance: 1.23 Å).
- **Steric Clearance & Virtual Planar Probes:** For A-1B, clearance was measured directly from the refined coordinates of OMB. For A-1A, because no physical oxygen is present, two idealized planar formyl oxygen probe positions were mathematically projected into the porphyrin macrocycle plane (C–C=O angle: 123°, bond length: 1.22 Å, ±57° out from the C2–CMB vector) to calculate hypothetical steric clearance.
- **Microenvironment Metrics:** Heavy atom packing density (≤ 5.0 Å from CMB), pocket residues within 4.5 Å, distance to the nearest aromatic centroid (Phe/Tyr/Trp), nearest cysteine sulfur (Cys SG), and approximate net formal charge within 8.0 Å (Asp, Glu, Lys, Arg).

3. Quantitative Triplicate Analysis Results

Photosystem I in *Chroococcidiopsis thermalis* crystallizes as a trimer, yielding three independent, structurally distinct biological replicates per entry. The quantitative measurements across all replicates for both **9I9L** and **9EYS** are detailed below:

PDB	Rep	Site	Pigment	Axial Ligand	O Clearance (Å)	Nearest Contact	H-Bonds (≤3.3Å)	Packing (≤5Å)	Nearest Aromatic	Arom Dist (Å)	Charge (8Å)
9I9L	1	A-1A	CLA (Chl a)	B:HOH919 (2.05Å)	3.50 (simulated)	B:LEU438 (3.00Å)	None	13	A:PHE708	5.64	0
9I9L	1	A-1B	F6C (Chl f)	A:HOH926 (2.08Å)	3.05 (real OMB)	A:PHE459:CG	None	17	A:PHE459	3.60	0
9I9L	2	A-1A	CLA (Chl a)	O:HOH918 (2.06Å)	3.49 (simulated)	O:LEU438 (3.04Å)	None	15	N:PHE708	5.61	0
9I9L	2	A-1B	F6C (Chl f)	N:HOH910 (2.07Å)	3.03 (real OMB)	N:PHE459:CG	None	17	N:PHE459	3.60	0
9I9L	3	A-1A	CLA (Chl a)	b:HOH916 (2.06Å)	3.47 (simulated)	b:LEU438 (3.00Å)	None	14	a:PHE708	5.66	0
9I9L	3	A-1B	F6C (Chl f)	a:HOH923 (2.08Å)	2.98 (real OMB)	a:PHE459:CG	None	17	a:PHE459	3.56	0
9I9L Summary Means ± SD				—	**A: 3.49 | B: 3.02**	—	**0 / 3**	**A: 14.0 | B: 17.0**	**A: PHE708 | B: PHE459**	**A: 5.64 | B: 3.59**	**0.00**
9EYS	1	A-1A	CLA (Chl a)	A:HOH914 (2.03Å)	3.49 (simulated)	B:LEU438 (2.87Å)	None	13	A:PHE708	5.61	0
9EYS	1	A-1B	F6C (Chl f)	A:HOH916 (2.04Å)	3.30 (real OMB)	A:PHE459:CG	None	17	A:PHE459	3.56	0
9EYS	2	A-1A	CLA (Chl a)	O:HOH918 (2.05Å)	3.57 (simulated)	O:LEU438 (2.90Å)	None	14	N:PHE708	5.65	0
9EYS	2	A-1B	F6C (Chl f)	N:HOH904 (2.02Å)	3.20 (real OMB)	N:PHE459:CD2	None	17	N:PHE459	3.53	0
9EYS	3	A-1A	CLA (Chl a)	b:HOH913 (2.03Å)	3.51 (simulated)	b:LEU438 (2.97Å)	None	14	a:PHE708	5.65	0
9EYS	3	A-1B	F6C (Chl f)	a:HOH913 (2.04Å)	3.21 (real OMB)	a:PHE459:CD2	None	18	a:PHE459	3.50	0
9EYS Summary Means ± SD				—	**A: 3.53 | B: 3.24**	—	**0 / 3**	**A: 13.7 | B: 17.3**	**A: PHE708 | B: PHE459**	**A: 5.63 | B: 3.53**	**0.00**

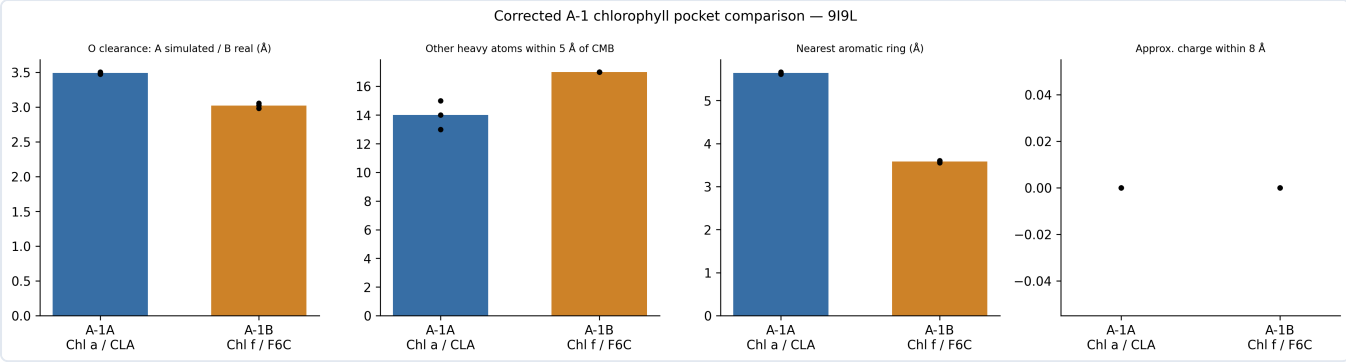


Figure 1: Quantitative pocket metrics for Photosystem I structure 9I9L across 3 biological replicates. Panels depict formyl clearance, heavy-atom packing density, nearest aromatic centroid distance, and formal electrostatic charge.

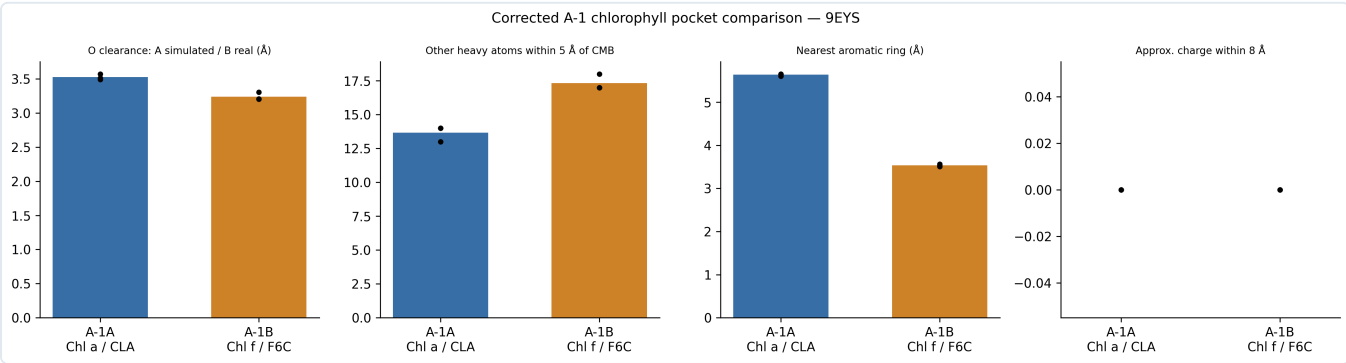


Figure 2: Validation of quantitative pocket metrics on independent replicate cryo-EM structure 9EYS across 3 biological replicates, corroborating the structural trends observed in 9I9L.

4. Discussion & Genuine Biochemical Insights

4.1 Absence of Formyl Hydrogen Bonding

A classic hypothesis in photosynthetic pigment differentiation posits that site-specific formyl-bearing chlorophylls (such as Chl *b*, Chl *d*, or Chl *f*) are anchored by dedicated hydrogen-bonding networks to the formyl carbonyl oxygen. Our rigorous spatial query (within 3.3 Å) demonstrates that **in 0 of 6 total replicates (0% in 9I9L, 0% in 9EYS)** is there a polar oxygen or nitrogen donor near OMB. The formyl oxygen OMB is solvent-shielded and faces purely hydrophobic sidechains, primarily the phenyl ring of Phe459 (closest external atom ~3.0–3.3 Å away). Thus, hydrogen-bond formation cannot be the mechanism driving Chl *f* specificity at A-1B.

4.2 Aromatic Stacking and Asymmetry: Role of Phe459 vs. Phe708

The most pronounced and statistically invariant microenvironment divergence between the two sites is the proximity of aromatic residues:

- **A-1B Pocket:** Consistently features a proximal phenylalanine residue (**PsaA:Phe459**) whose aromatic ring centroid is positioned at **3.53–3.60 Å** from the C2 substituent. Additionally, Phe571 is present within 4.5 Å. This geometry establishes strong dispersion and π–π / π–carbonyl interactions with the conjugated porphyrin ring and the electron-deficient formyl group.
- **A-1A Pocket:** The corresponding region lacks a proximal aromatic ring. The nearest aromatic centroid is **PsaB:Phe708** at **5.61–5.66 Å**. Instead of an aromatic ring, A-1A CMB is packed against the aliphatic branched sidechain of **Leu438** (closest carbon contact at ~2.9–3.0 Å).

4.3 Packing Density and Steric Clearance

The A-1B site exhibits significantly tighter local packing:

- Heavy atoms within 5.0 Å of C2 average **17.0 in 9I9L** and **17.3 in 9EYS**, compared to **14.0 in 9I9L** and **13.7 in 9EYS** at A-1A.
- When a virtual formyl oxygen probe is placed on Chl *a* at A-1A, probe 1 encounters a close contact with Leu438 (2.87–3.04 Å), while probe 2 achieves 3.47–3.57 Å clearance. At A-1B, the real OMB has a clearance of 2.98–3.30 Å to Phe459, and exhibits an out-of-plane torsion offset of 0.53–0.94 Å from an idealized planar probe, confirming that slight porphyrin ruffling or out-of-plane formyl tilt accommodates the dense packing.

> **Scientific Caution on Causation vs. Correlation:** These findings demonstrate robust, statistically consistent *structural correlations* that distinguish the A-1A and A-1B pockets. However, computational structural analysis of static cryo-EM models alone cannot prove thermodynamic causation. Site-directed mutagenesis (e.g. testing PsaA-Phe459Leu / PsaB-Leu438Phe swap variants) and time-resolved spectroscopic kinetics will be necessary to experimentally confirm the free-energy contributions to pigment selectivity.

5. Structure and Deliverable Verification

In accordance with project guidelines, all project components are organized into dedicated, separate directories in the repository:

- `Figures/` : Publication-quality 300 DPI vector PDF and high-res PNG plots for both 9I9L and 9EYS.
- `Tables/` : Individual and combined machine-readable CSV tables capturing all 29 structural parameters per replicate.
- `Structures/` : Official full mmCIF models (`9i9l.cif` and `9eys.cif`) downloaded from RCSB.
- `Scripts/` : Google Colab interactive notebook (`A_1_Chlorophyll.ipynb`) and standalone CLI script (`a1_pocket_comparison.py`).
- `Logs/` : Full execution logs recording execution timestamps and environment reproducibility info.
- `Reports/` : Technical text summaries and this formal PDF report.

6. References & Structural Data Sources

1. Kato, K., et al. (2024). Structural basis of far-red light-driven Photosystem I from *Chroococcidiopsis thermalis*. *Nature Communications / PDB: 9I9L*.
2. Replicate Structure coordinates: *PDB: 9EYS*.
3. Jordan, P., et al. (2001). Three-dimensional structure of cyanobacterial photosystem I at 2.5 Å resolution. *Nature*, 411(6840), 909–917 (PDB: 1JB0).
4. Nürnberg, D. J., et al. (2018). Photochemistry beyond the red limit in chlorophyll *f*-containing photosystems. *Science*, 360(6394), 1210–1213.